APPENDIX J2

PROVISION FOR COMPUTING AND NETWORKING FACILITIES FOR THE *PROJECT MANAGER* AND THE *SUPERVISOR*

[PS Clauses 1.49]

A. GENERAL

1 Computing and networking facilities shall be provided for exclusive use of the *Project Manager* and the *Supervisor* at the *Project Manager*'s Site Accommodation to perform work, communicate, share and transfer data with other offices. Such facilities shall be separated from facilities used by the *Contractor*. The facilities supplied shall include structured computer cabling system with maintenance and support services thereto.

2 The *Contractor* shall subscribe Internet service with Internet service provider(s) such that the computer networks in the *Project Manager*'s Site Accommodation can be connected to the Internet and no time-out set for the connection. The speed of the Internet service shall be 100Mbps downstream and 50Mbps upstream minimum. The *Contractor* shall liaise with the internet service provider(s), the public telecommunication line carriers and its subcontractors, if necessary, to provide on-site Broadband Equipment setup with modem and/or IP sharer on loan and router configuration to establish the Internet connection for the *Project Manager*'s Site Accommodation.

3 The *Contractor* shall obtain the *Supervisor*'s consent on the plan and speed of the telecommunication line(s) to be installed prior to installation.

4 The *Contractor* shall propose and pay for an independent service provider, which does not have close commercial relation with the *Contractor*, to provide maintenance and support services for the entire computer facilities, including its network, in the *Project Manager*'s Site Accommodation for the duration of the contract subject to the acceptance of the *Supervisor*. The *Contractor* shall notify the *Supervisor* of the name of the service provider.

5 The support services from the independent service provider shall include administration of the local area network, equipment installation and configuration with at least 1 workstation with network security firewall, DMZ, intrusion detection system, routine maintenance of the equipment, and ad hoc user support as requested by the *Project Manager*'s and the *Supervisor*'s staff. The system support services shall not involve any staff from the *Contractor*.

6 The response time for providing support services in the above paragraphs 4 and 5 shall be within 24 hours on request by the *Project Manager*'s and the *Supervisor*'s staff.

7 The contract computer facilities shall comprise items specified in "Schedule of Data Processing Equipment for the *Project Manager* and the *Supervisor*".

8 Within 14 days of the Contract Date, or such other period as specified in the contract, the *Contractor* shall submit for the *Supervisor*'s acceptance details of the contract computer facilities to be provided. The proposed equipment shall be well proven and shall not infringe any copyrights.

9 Within 14 days of receiving the *Supervisor*'s acceptance or other such date accepted by the *Supervisor*, the *Contractor* shall deliver the contract computer facilities to the *Project Manager*'s Site Accommodation, or to location as specified by the *Supervisor*. All software shall be delivered to the location specified by the *Supervisor*, in intact shipping cartons with all seals unbroken. At the time of delivery, the originals of all warranty agreements, invoices and other supporting documents related to the price of the equipment shall be produced for inspection by the *Supervisor*.

10 Software included in the contract computer facilities shall be licensed in the name of the Government of HKSAR. The *Contractor*'s attention is drawn to the Intellectual Property (Miscellaneous Amendments) Ordinance 2000 that all software provided must have proper licenses. The *Contractor* shall adopt the following software asset management (SAM) measures to ensure that only properly licensed software is used:

(a) appoint a compliance officer responsible for managing software assets;

(b) promulgate internal instruction in respect of the proper management and use of computer software to ensure each site staff clearly understands the requirements;

(c) maintain and implement proper procedures for installing and distributing software;

(d) conduct periodic software asset audit;

(e) confirm proper licensing and authorisation for all software used.

11 Within five working days of delivery, the *Contractor* shall arrange for the contract computer facilities as specified in paragraph 7 to be installed and tested to complete working order to the acceptance of the *Supervisor*.

12 The *Contractor* shall license and maintain the equipment until the expiry of the *defects date*, or where there is more than one such *defects date*, the expiry of the latest *defects date* or other agreed date if so directed by the *Project Manager*.

13 All systems and equipment shall be configured and managed to comply with the following regulations and policy of HKSARG: HyD Departmental IT Security Policy, Baseline IT Security Policy (S17), IT Security Guideline (G3), Internet Gateway Security Guideline (G50) and other relevant guidelines from OGCIO through the following URL below:

https://www.ogcio.gov.hk/en/our_work/information_cyber_security/government/

14 The *Contractor* shall maintain all the computing & networking facilities and associated devices in the *Project Manager*'s Site Accommodation.

15 The *Contractor* shall be required to transport and relocate the computing, networking facilities and associated devices to other contractor's office until the *defect* date.

B. SCHEDULE OF DATA PROCESSING EQUIPMENT FOR THE SUPERVISOR

1. DESKTOP MICROCOMPUTERS

1.1 Type I - Microcomputer for General Purposes

The microcomputer shall comply with the minimum technical specifications listed below and those under Section 1.2 of this Appendix.

System Unit
· Processors: Intel® Core™ i5-10500 6 Core Processor (3.1GHz & 12MB Cache), equivalent or better performance; · Memory: At least 8GB RAM; · Hard Disk: At least 2TB SATA 7200rpm Hard Drive;

· Chipset: Intel H470, equivalent or better performance; · DVD RW/DL: At least 8x with burning Software; · Expansion Slots: At least (1x PCIe x16, 1x PCIe x1); · Form Factor: Small Form design; · Integrated High Definition Audio, with internal speaker; · BIOS with built-in password control; · Power on self-testing of system component and reset facilitates protected from inadvertent operation; · Power Supply: Up to 200W; 80 PLUS; · ENERGY STAR Qualified and RoHS Compliant.
Video Card
· Integrated Intel HD Graphics or equivalent, At least 1x VGA, 2x DisplayPort output; · Supporting true colour non-interlaced of 60Hz vertical scan rate at 1920x1080 resolution, supporting full screen display on interlaced and non-interlaced monitors; · Synchronizing up in all aspects with the best performance of the monitor provided; · Enhanced driver support for Windows 10 and above.
Communication Ports
· Minimum 6x USB 3.0 Ports and 4x USB2.0 Ports (at least 2x USB 3.0 Ports at the front panel) or better performance; · 1x Microphone/Headphone Combo; · 1x at least 100Mbps Ethernet Port; · 1x Audio Line-in/1x Microphone-in, 1x Audio Line-out.
Keyboard
· USB Typewriter-like enhanced 101-keyboard IBM compatible; good rebounding and pleasant tactile and click feel, detachable; Traditional Chinese version; · Not less than 12 programmable function key with upper and lower case, provided with screen dump key.
Others
· Optical USB Mouse (2 buttons with scrolling function) or compatible pointing device with user definable buttons; · USB Chinese handwriting recognition input device; · Management agents, Rom-based setup, Quick restore solution, Intelligent Manageability: Pre-failure warranty; · Security: Diskette write & boot control, Flash ROM lack, keyboard cable lock provision, power-on password, parallel & serial interface control, cable lock provision & chassis intrusion alert; · 1x Power Cable & Recovery Media; · UTP Cable: (i) Category 5e or better performance UTP Cable; (ii) Cabling (UTP) Service Installation at Site Office (including all necessary accessories & cable).
Documentation
· The *Contractor* shall supply sufficient documents or manuals as shown in the

following for each computer set to the *Supervisor*: (i) Manuals on how to operate the computer set; (ii) Manuals on how to use the facilities and software provided by the supplier / the Contractor.
Warranty
· 3-Year free on-site.
Software
· Preload Windows 11 Professional 64-bit (with Chinese language package) which support the option of downgraded installation as Windows 10 Enterprise; · Windows 11 Enterprise E3 64-Bit LTSC upgrade and Software Assurance License (with Chinese language package); · Microsoft Office Standard 2019 (Chinese) 32-bit or latest version; · Trend Micro Apex One Security Agent · Adobe Acrobat Professional XI English or Traditional Chinese for Windows 10 or above · Cisco Umbrella Insights · Management agents, Rom-based setup, Quick restore CD.
Monitor
· Panel Size & Type: 23.8” or above LCD or LED; · Aspect ratio: 16:9 with auto picture format to 4:3 mode; · Optimum Resolution: 1920 x 1080 @60Hz or above; · Display Colors: 16.7M display colors or better; · Response Time: 5ms or less (Typical); · Contrast Ratio: 1000:1 (Typical); · Brightness: 250 cd/m^2 (Typical) or above; · Viewing angle: At least 178° (H) /178° (V); · Stand: Tilt, Swivel -170/+170 degree and Height adjustment 130mm, Pivot; · Audio: Built-in speakers; · Special Features: with Blue Light Filter & Flicker Free; · Connectivity: At least 1x Display Port, 1x HDMI, 1x Audio Line-in, 1x Headphone; · 1x Power Cable, 1x DisplayPort Cables (2m); · ENERGY STAR Qualified.
Warranty
· 3-Year free on-site.

1.2 Technical Specification applicable to Type I Microcomputers

The microcomputer shall comply with the technical specification listed below:

a. Functional Requirement

The microcomputer shall be certified to operate under current version of English and traditional Chinese Microsoft Windows 10 Enterprise LTSC 1809, Compatible.

b. CPU, Motherboard and Memory

(i) PCI and PCI Express bus architecture system mother board with an onboard integrated graphic display port;

(ii) Power on self-testing of system component and reset facilitates protected from in adverted operation;

(iii) real time clock and calendar;

(iv) security key lock switch and power reset switch;

(v) power on self-testing of system component and reset facilities protected from inadvertent operation;

(vi) BIOS with built-in password control;

(vii) no less than 2 free PCIe expansion slots.

(viii) Integrated High Definition Audio, with internal speaker;

(ix) Flexible positioning chassis (not larger than: 40cm(H) x 12cm(W) x 45cm(D)); Tool-less chassis design; and

(x) ENERGY STAR and RoHS compliant.

c. Secondary Storage

(i) Serial ATA Revision 3.0 interface for minimum 2 hard disks;

(ii) USB 3.0 Hi-Speed Media Card Reader, supporting CF, SD, SDHC, SDXC & micro-SD cards; and

(iii) one internal 16X DVD-ROM + 52X CD-RW combo drive or better with burning Software.

d. Display Adapter (unless otherwise specified)

(i) Integrated Intel® HD Graphics 2000 or equivalent, with VGA, DVI or DisplayPort output;

(ii) Supporting true colour non-interlaced of 75Hz vertical scan rate at 1280×1024 resolution, supporting full screen display on interlaced and non-interlaced monitors;

(iii) Synchronizing up in all aspects with the best performance of the monitor provided; and

(iv) Enhanced driver support for 32-bit Windows 10 and above.

e. Keyboard

(i) typewriter-like enhanced 101-keyboard IBM compatible; good rebounding and pleasant tactile and click feel, detachable; Traditional Chinese version; and

(ii) not less than 12 programmable function keys with upper and lower case, provided with screen dump key.

f. Communication Ports

(i) minimum six USB 2.0 ports and four 3.0 ports.

(ii) 1 no. PCI 10/100/1000 MB Ethernet adapter card or an onboard integrated 100/1000 Ethernet port.

(iii) UTP Cable

1) Category 5e UTP Cable; and

2) Cabling (UTP) Service Installation at Site Office (including all necessary accessories & cable)

g. Mouse

Two-button or three-button USB type optical mouse.

h. Power Supply Unit

The power supply unit for the computer shall be of sufficient capacity to provide power supply to all components connected with the computer including allowance for future expansion cards to be inserted in all unused expansion slots. The unit shall have at least four number of device connectors for internal storage devices without using power splitters or tee-off cables.

i. Documentation

The *Contractor* shall supply sufficient documents or manuals as shown in the following for each computer set to the *Supervisor*:

(i) Manuals on how to operate the computer set; and

(ii) Manuals on how to use the facilities and software provided by the supplier / the *Contractor*;

j. Others

(i) Bundled with management agents, Rom-based setup, Quick restore solution; Intelligent Manageability: Pre-failure warranty;

Security: Diskette write & boot control, Flash ROM lack, keyboard cable lock provision, power-on password, parallel & serial interface control, cable lock provision, and chassis intrusion alert; and

(ii) Bundled with connecting cables & power cord.

(iii) Bundles with all hardware drivers support for Windows 11 and above.

2. NOTEBOOK MICROCOMPUTER (PRESENTATION)

The notebook microcomputer shall be certified to operate under current version of English and traditional Chinese Windows® 10 Professional.

Notebook computers shall comply with the minimum technical specification listed below:

System Unit
· Processors: Intel® Core™ i5-10210U 4 Core Processor (1.6GHz & 6MB Cache), equivalent or better performance; · Memory: At least 16GB RAM; · Hard Disk: At least 512GB SSD; · Integrated High Definition Audio, with internal speaker; · BIOS with built-in password control; · Power on self-testing of system component and reset facilitates protected from inadvertent operation; · ENERGY STAR Qualified and RoHS Compliant.
Video Card
· Integrated Intel HD Graphics or equivalent, 1x DisplayPort output; · Enhanced driver support for Windows 10 and above.
Display Size
· Not less than 14” (diagonal)
Communication Ports
· Minimum 1 x USB 3.0 Ports and 2 x USB2.0 Ports; · 1 x Microphone/Headphone Combo; · 1 x 100Mbps Ethernet Port or better;
Media Card Reader
· Supports SD, SDHC, SDXC
Wireless Technologies
· 802.11b/g/n/ac and Bluetooth® 5.0 or above
Keyboard
· Full-size island-style, spill-resistant keyboard
Point Devices
· Touch pad with multi-touch gesture support
Camera
· Min. 720p HD Camera
Battery life
· Min. 10 hours
Weight
· Not exceeding 2.3kg in weight including the internal rechargeable battery
Documentation
· The Contractor shall supply sufficient documents or manuals as shown in the following for each computer set to the Engineer: (i) Manuals on how to operate the notebook; (ii) Manuals on how to use the facilities and software provided by the supplier / the Contractor.
Warranty
· 3-Year free on-site.
Software
· Preload Windows 11 Professional 64-bit (with Chinese language package) which

support the option of downgraded installation as Windows 10 Enterprise;

- Windows 11 Enterprise E3 64-Bit LTSC upgrade and Software Assurance License (with Chinese language package);
- Microsoft Office Standard 2019 (Chinese) 32-bit or latest version;
- Trend Micro Apex One Security Agent
- Adobe Acrobat Professional XI English or Traditional Chinese for Windows 10 or above
- Cisco Umbrella Insights
- Management agents, Rom-based setup, Quick restore CD.

3. A3 /A4 COLOR NETWORK MULTIFUNCTION PRINTER (RICOH IM C6000 – ANNUAL METER CHARGES)

The A3/A4 Color Network Multifunction Printer (Ricoh IM C6000) shall comply with the following specifications:

Printer	
Print speed	Up to 60ppm for color/mono
Resolution	Up to 1200 x 1200 dpi
Memory	At least 1,024MB memory
First Page out	Less than 5 sec. for color/mono
Printer language	PCL 5c, PCL 6, PostScript 3 emulation, PDF Direct emulation
Input Capacity	1 x bypass tray, + 1 x 500-sheets tray
Output Capacity	Min. 250 sheets
Interface	10/100/1000 Base-TX Ethernet, USB 2.0
Media Size	SRA3, A3, A4, A5, A6, B4, B5, B6
Media Type	Envelope/ Labels/ Card stock/ Plain paper
Media Handling	Auto Duplex printing
Scanner	
Scan Speed	120ipm (Simplex) / 240ipm (duplex)
Resolution	1,200 dpi

Compression Method	TIFF (MH, MR, MMR, JBIG2), Grayscale
Copier	
Multiple Copying	Up to 999 Copies
Resolution	600 dpi
Others	Comply with Energy Star, RoHS, power saving operation mode with printer management software
Warranty	3-Year free on-site.

4. CHINESE HANDWRITER

The Chinese handwriter shall be PenPower 5” x 4” Hand Writer or products having equivalent functions and performance complying with the following specifications:

a. Writing area shall not be less than 5” x 4”;

b. USB interface;

c. The handwriter shall be provided with drivers for current release of English and traditional Chinese versions of Windows® 10or above;

d. The handwriter shall be compatible with the latest version of Microsoft Office.

5. NETWORK EQUIPMENT.

a. Cable shall be threaded through conduits from ISP Switch to each room and terminated at information outlets at positions as agreed by the *Supervisor*;

b. The *Contractor* shall design for the cable routes and submit the proposal for the *Supervisor*’s approval prior installation;

c. Data cable shall be unshielded twisted pair (UTP) cable supporting IEEE 802.3 1G/10G Base-T standards and shall comply with the following requirements:

Category 5eCable Requirement

	CAT5e
Frequency	100 MHz
Attenuation (min. at 100 MHz)	22 dB
Characteristic Impedance	100 ohms = 15%
NEXT (min. at 100 MHz)	35.3 dB
PS-NEXT (min. at 100 MHz)	32.3 dB

EL-FEXT (min. at 100 MHz)	23.8 dB
PS-ELFEXT (min. at 100 MHz)	20.8 dB
PS-ANEXT (min. at 500 MHz)	--
PS-AELFEXT (min. at 500 MHz)	20.1 dB
Return Loss (min. at 100 MHz)	20.1 dB
Delay Skew (max. per 100m)	45 ns
Networks Supported	1000BASE-T

d. The information outlet shall be a dual gang shuttered faceplate equipped with two RJ45 jacks which shall be of IDC type as specified in TIA/EIA 568-B / TIA/EIA 568-C

e. The information outlet shall be able to intercept two UTP cables as well as connection of workstation and / or terminal via UTP patch cords.

f. Network switches shall support SNMP protocol or products having equivalent function or performance.

g. The *Contractor* shall liaise with the internet service provider(s), the public telecommunication line carriers and its subcontractors to ensure the telecommunication line(s) connection can be installed inside ISP Switch to ensure that connectivity can be established for the *Project Manager*'s Site Accommodation.

h. In order to prevent illegal access to the computer facilities from outside, a hardware firewall shall be provided to check and control access to local area network. The hardware firewall shall be16-Port Firewall Switch or products having equivalent functions and performance with prior acceptance from the *Supervisor*.

i. The LAN shall be capable of integrating the various equipment and operating systems stated in this Appendix. The network shall offer an extensive open architecture and shall be easily reconfigured and restructured to cater for changes in computing requirements. The *Contractor* shall provide the maintenance service of the LAN (software and hardware) over the entire contract period.

j. The *Contractor* shall provide all necessary equipment for the LAN including Hubs, Switch and Router.

6. SYSTEM INTEGRATION AND COMMISSIONING

a. The *Contractor* shall provide a detailed as-built document of the computer facilities including the entire network topology and the inventory list of the computer hardware and software.

7. SOFTWARE REQUIREMENT

The following software shall be provided and installed in the computers as specified in below. The software shall comply with the following specifications.

7.1 Operating system

a. Type I and notebook microcomputers

(i) English or traditional Chinese version (as directed by the *Supervisor*) Windows 10 Enterprise LTSC 1809

7.2 Terminal licenses for the application software packages

a. Type I and notebook microcomputers

(i) ESET NOD32 Antivirus or products having equivalent functions and performance (total of 8 user licenses);

(ii) Microsoft Office Standard 2019 Edition or latest version 32 bit (as directed by the *Supervisor*) (with Microsoft Word, Microsoft Excel, Microsoft Outlook, Microsoft PowerPoint) (total of 8 user licenses); and

(iii) Adobe Acrobat Professional XI English or Traditional Chinese for Windows OS for Windows OS or equivalent (total of 8 user licenses).

8. ENVIRONMENTAL CONDITION

8.1 The computer system must be able to function properly and reliably under office environmental conditions defined as follows:

Temperature: 10 to 30 degrees Centigrade

Humidity: 20% to 80% (non-condensing)

8.2 Power supply for the system shall be 220 volts AC, 50Hz from normal building wiring circuit mains. Power regulator, stabilizer or transformer shall be supplied by the Contractor for the computer system such that the system can function efficiently.

9. DOCUMENTATION

9.1 Sufficient documentation must be supplied with the system. The documentation shall be self-tutorial in nature and be readily understood by non-computer personnel.

9.2 The following manuals must be supplied with the system:

a. Manual on how to operate the equipment, and

b. Manual on how to use the facilities and software provided by the supplier.

10. TRAINING

10.1 Unless otherwise instructed by the *Supervisor*, the *Contractor* shall provide training for two persons for the following:

a. Hardware operation;

b. Use of the software specified in Section 7 herein;

c. Administration of the local area networks.

10.2 Training centre recommended by the *Contractor* should be from the current list of the Government Microcomputer-Related Training Contracts.

11. DATA MODEM AND INTERNET SERVICES CONNECTION

A modem of data transfer speed of minimum downstream speed of 100Mbps and upstream speed of 50Mbps respectively. One Broadband Internet service supporting the internet access (including all hardware and/or software supporting networking security eg. Firewall) and continuously providing all expenses on installation; wiring, maintenance and all necessary license and fees up end of the *defects date*.

Broadband Internet Services	- Minimum downstream speed of 100Mbps and upstream speed of 50Mbps and from a government approved service provider - On-site Broadband Equipment setup (modem and/or IP sharer on load) including router configuration - Broadband configuration with network security firewall - Unlimited internet access - No time-out set for the connection
Network Security	The designated desktop computers to be installed with Internet must not be connected into the Local Area Network and a bandwidth sharing device shall be provided by the Contractor to enable sharing of internet access between computers. Personal firewall shall be required for individual desktop computer

12. INSTALLATION

The data processing equipment shall be installed in the accommodation of the *Supervisor* as specified in the following tables.

Items	**Quantities**
Type I Microcomputer Workstation	8
Notebook microcomputers	1
A3/A4 Color Network Multifunction Printer (Ricoh IM C6000 – Annual Meter Charges)	1
Chinese Hand Writer	2

13. SECURITY

13.1 All systems and equipment shall be configured and managed to comply with the following regulations and policy of HKSARG: HyD Departmental IT Security Policy, Baseline IT Security Policy (S17), IT Security Guideline (G3), Internet Gateway Security Guideline (G50) and other relevant guidelines from OGCIO through the following URL below:

https://www.ogcio.gov.hk/en/our_work/information_cyber_security/government/